



Deep learning and vision transformers-based framework for breast cancer and subtype identification

Ishrat Jahan¹ · Muhammad E. H. Chowdhury² · Semir Vranic³ · Rafif Mahmood Al Saady³ · Saidul Kabir⁴ · Zahid Hasan Pranto¹ · Sabiha Jahan Mim¹ · Sadia Farhana Nobil²

Received: 25 August 2024 / Accepted: 3 January 2025 / Published online: 29 January 2025
© The Author(s) 2025, corrected publication 2025

Abstract

Breast cancer, marked by uncontrolled cell growth in breast tissue, is the most common cancer among women and a second-leading cause of cancer-related deaths. Among its types, ductal and lobular carcinomas are the most prevalent, with invasive ductal carcinoma accounting for about 70–80% of cases and invasive lobular carcinoma for about 10–15%. Accurate identification is crucial for effective treatment but can be time-consuming and prone to interobserver variability. AI can rapidly analyze pathological images, providing precise, cost-effective identification, thus reducing the pathologists' workload. This study utilizes a deep learning framework for advanced, automatic breast cancer detection and subtype identification. The framework comprises three key components: detecting cancerous patches, identifying cancer subtypes (ductal and lobular carcinoma), and predicting patient-level outcomes from whole slide images (WSI). The validation process includes visualization using Score-CAM to highlight cancer-affected areas prominently. Datasets include 111 WSIs (85 malignant from the Warwick HER2 dataset and 26 benign from pathologists). For subtype detection, there are 57 ductal and 8 lobular carcinoma cases. A total of 28,428 annotated patches were reviewed by two expert pathologists. Four pre-trained models—DenseNet-201, MobileNetV2, an ensemble of these two, and a Vision Transformer-based model—were fine-tuned and tested on the patches. Patient-level results were predicted using a majority voting technique based on the percentage of each patch type in the WSI. The Vision Transformer-based model outperformed other models in patch classification, achieving an accuracy of 96.74% for cancerous patch detection and 89.78% for cancer subtype classification. For WSI-based cancer classification, the majority voting method attained an F1-score of 99.06 and 96.13% for WSI-based cancer subtype classification. The proposed deep learning-based framework for advanced breast cancer detection and subtype identification yielded promising results. This advanced framework shows great promise in medical practice, offering an economical, efficient solution for generating accurate, clinically relevant results and enhancing diagnostic accuracy in hospitals, research centers, and pathology laboratories. Nonetheless, further studies are needed to validate its effectiveness across various environments and larger datasets.

Keywords Breast cancer · Histopathology image · Hematoxylin–eosin (H&E) · Ductal carcinoma · Lobular carcinoma · Deep learning · Vision Transformer

1 Introduction

Cancer is a disease characterized by uncontrolled cell division, leading to the proliferation of abnormal cells that invade and damage surrounding healthy tissues [56]. This process may give rise to tumors, damage the immune system, and cause other harmful effects, potentially leading to fatal outcomes. Normally, cell proliferation replaces diseased cells. Uncontrolled replication results in tissue or

cell masses, forming tumors or cancers such as lung, liver, colorectal, stomach, and breast cancers [1]. Breast cancer begins in the breast with a cluster of abnormal cells that can grow out of control, pushing into nearby tissues or traveling to other parts of the body. It originates from the epithelial cells lining the ducts or lobules within the glandular tissue of the breast [18]. According to the National Center for Health Statistics, cancer ranks as the second-leading cause of human mortality. In 2023, an estimated 297,790 women and 2800 men in the United States will be diagnosed with invasive breast cancer [45].

Extended author information available on the last page of the article

There are two primary types of breast tumors: benign and malignant. In contrast to benign, malignant cells proliferate rapidly due to their irregular appearance and structure, leading to swift division and potential spread throughout the body [35]. These tumors have the potential to metastasize, meaning they can detach and travel through blood or lymph vessels to distant sites, causing damage to other organs and tissues. Breast cancer comes in two main types: in situ (not spreading) and invasive. In situ cancers stay confined to certain areas, like ducts, while invasive ones can spread further [40]. Ductal carcinoma in situ (DCIS) is when abnormal cells are found inside ducts but have not spread. Lobular carcinoma in situ (LCIS) is not actually cancer, but it increases the risk of developing cancer in future [8]. Invasive breast cancers, unlike their in situ counterparts, exhibit a crucial distinction: They breach the protective basement membrane, allowing malignant cells to infiltrate surrounding healthy tissues [7]. This signifies a more advanced stage of the disease and underscores the importance of early detection. The most prevalent type, invasive ductal carcinoma (IDC), accounts for a significant portion (70–80%) of breast cancer diagnoses [3, 19]. It originates in the cells lining the milk ducts, potentially disrupting their normal function and hindering milk flow. Invasive lobular carcinoma (ILC), although less frequent (around 10% of cases), arises in the lobules responsible for milk production [11]. ILC is the most common special type of breast cancer and often metastasizes unpredictably to rare locations like the gastrointestinal tract, ovaries, and leptomeninges. Several studies indicate that it has a worse clinical outcome than ductal carcinoma [38]. Risk factors include genetic mutations, hormonal influences, aging, and family history. Symptoms may involve breast lumps, skin changes, nipple discharge, and discomfort.

When breast cancer is identified in its initial stages, it can be treated effectively before it has the chance to metastasize to other areas of the body [16]. The National Breast Cancer Foundation (NBCF) has reported that early detection of breast cancer (BCD) increases the potential for successful treatment. A variety of imaging methods, including thermography, mammography, ultrasound, and pathology testing, are used to identify breast cancer [32, 44]. Because of the elevated mortality associated with breast cancer, women are advised to undergo routine screenings, which typically involve mammograms and computerized tomography (CT) scans [12]. However, these noninvasive methods may not adequately capture the diverse behaviors exhibited by breast tumors. Consequently, pathological analysis is conducted as a standard reference to gain insight into the pathophysiology of breast tumors. A biopsy is an invasive procedure that requires pathologists to examine the tissue sample and determine

the nature of a breast abnormality. Typically, the obtained biopsy is stained by hematoxylin and eosin (H&E) staining. Hematoxylin binds to deoxyribonucleic acid (DNA), resulting in purple or blue staining of cell nuclei, while eosin interacts with proteins, producing a pink stain in other cellular structures [2]. Subsequently, pathologists examine the slides under a microscope to determine a potential diagnosis of breast cancer [54]. To diagnose any type of cancer, including breast cancer, a histological imaging diagnosis is the gold standard [15, 26, 57]. Currently, histopathological image examination relies on manual assessment by pathologists. However, this approach presents several potential issues, such as resource shortages in certain regions or medical facilities where pathologists may be unavailable. This shortage of resources could result in a higher number of cancer cases not being examined in their early stages. Failure to detect breast cancer in its early stages may result in the spread of cancer cells to other areas of the body, potentially causing harm in those locations as well. Nearly, 75% of National Health Services (NHS) trusts and 76% of health boards have reported insufficient medical specialists to effectively treat individuals with breast cancer. Even with a pathologist present to review histopathological images, the accuracy of the results hinges on the pathologist's expertise, diagnostic acumen, and comprehensive understanding of the field. Due to these challenges, there is a need for efficient and precise automated techniques for the detection and classification of breast cancer. Over the past decade, various machine learning techniques have been utilized for the classification of breast cancer based on histopathological images [20, 24, 47]. Motivated by the recent progress in machine learning and image processing, researchers have demonstrated a considerable interest in the development of computer-aided diagnostic systems for breast cancer [24].

The integration of AI in medical imaging for cancer detection has led to several advancements in model design and training strategies. Li et al. [28] introduced the U-KAN model, a strong backbone architecture designed for enhanced feature extraction in medical image segmentation and generation tasks, proving effective in capturing complex details. Yang et al. [58] proposed masked relation modeling (MRM), which incorporates genetic data into pre-training, improving the accuracy of medical image analysis by learning nuanced relational patterns. To address dataset variability, Li et al. [27] explored domain generalization using episodic training and task augmentation, increasing adaptability across diverse imaging datasets. Sun et al. [50] developed a few-shot learning method based on a global correlation network with discriminative embedding, which achieves precise segmentation even with limited data availability. Hirra et al. [20] adopted a patch-based deep learning framework for breast cancer

classification, using unsupervised pre-training and supervised fine-tuning to capture relevant features. Their model classifies overlapping 32×32 pixel patches as benign or malignant based on a 70% abnormal pixel threshold, with logistic regression used to classify the entire image. However, the framework achieves a relatively modest accuracy of 86%, indicating potential limitations in classification performance. Li et al. [29] presented a patch selection method for breast cancer classification that combines a clustering algorithm with a convolutional neural network (CNN). Applied to a 4-class histology image dataset, this method achieved 95% accuracy on the initial test set but showed an overall accuracy of 88.89%. Nonetheless, this approach lacks whole-slide image (WSI) level classification, crucial for patient-level diagnosis, as patch-based results reflect only localized regions. As a result, it falls short of the comprehensive accuracy needed for clinical standards. Mi et al. [31] utilized a two-stage architecture combining deep learning and machine learning for multi-class classification of breast pathology images, categorizing samples into normal tissue, benign lesions, ductal carcinoma in situ, and invasive carcinoma. Even though they applied their method to two datasets (BreakHis and BACH) and addressed multiclass classification, the study lacks visualization outputs, which are essential for interpreting model decisions in clinical settings. Additionally, the achieved accuracy of 86.67% at the patch level falls short of the reliability needed for clinical use. Nazeri et al. [33] proposed a patch-wise network functioning as an auto-encoder to capture features from image patches, followed by an image-wise network for whole-image classification. This approach combines locally focused and globally focused networks, trained on the ICIAR 2018 BACH dataset, and achieved a validation accuracy of 95%. However, their method lacks visualization techniques that would support interpretation and validation of the model's results. Obayya et al. [34] proposed the AOADL-HBCC technique for breast cancer classification using histopathology images, achieving 96.77% accuracy with a combination of AOA, SqueezeNet, and a DBN classifier. However, their approach is limited to binary classification. Srikantamurthy et al. [49] developed a hybrid CNN-LSTM model for classifying breast cancer histopathology images into benign and malignant subtypes, achieving 98% accuracy for binary classification at $400 \times$ magnification and 92.5% for multi-class subtype classification. The model used transfer learning on ImageNet and was evaluated on the BreakHis dataset across multiple magnifications (40x, 100x, 200x, and 400x). Limitations include the lack of whole-slide image (WSI) classification for comprehensive patient-level diagnosis and the absence of visualization techniques for model interpretability.

To address the limitations observed in previous studies, we introduce a comprehensive deep-learning framework designed to classify benign and malignant tissue at both patch-level and whole slide level, as well as distinguish between ductal and lobular carcinoma at the same levels. Our methodology involves a rigorous fivefold cross-validation approach, splitting the dataset into training, testing, and validation sets. Additionally, we apply data augmentation techniques to the training set, ensuring the development of a robust and reliable model. Furthermore, we incorporate advanced visualization techniques, such as class activation mapping, to highlight relevant regions within tissue samples, enhancing interpretability and enabling thorough validation of the model's decision-making process. The model's performance is meticulously evaluated to validate its efficacy. The key contributions of this study are listed below:

- Comprehensive experimental results are presented for benign vs. malignant and ductal vs. lobular classification at both patch-wise and patient-wise levels, providing valuable insights for cancer detection and subtype identification.
- Unlike many existing studies that utilize higher magnification levels (40x, 100x, 200x, 400x), experiments were conducted at lower magnification levels (10x and 20x). This approach maintained the integrity of the analysis while substantially reducing computational expenses, making the method more accessible and efficient.
- Score-CAM (Score-Weighted Class Activation Mapping) visualization techniques were employed to provide interpretability to the deep learning model [51]. This visualization highlighted the most relevant regions in the histopathological images that contribute to the classification [55], enhancing transparency in the model's decision-making process and providing valuable insights for pathologists.

The subsequent sections of the paper are organized as follows: In Sect. 2, an in-depth discussion of the materials and methods utilized in this study is provided, encompassing details regarding the dataset employed and the deep learning models implemented. Section 3 offers a comprehensive analysis of the quantitative and qualitative performance of the proposed framework. In Sect. 4, we present a discussion, including a comparison with earlier studies. Finally, in Sect. 5, we conclude by summarizing the key findings and outlining potential directions for future research.

2 Materials and methods

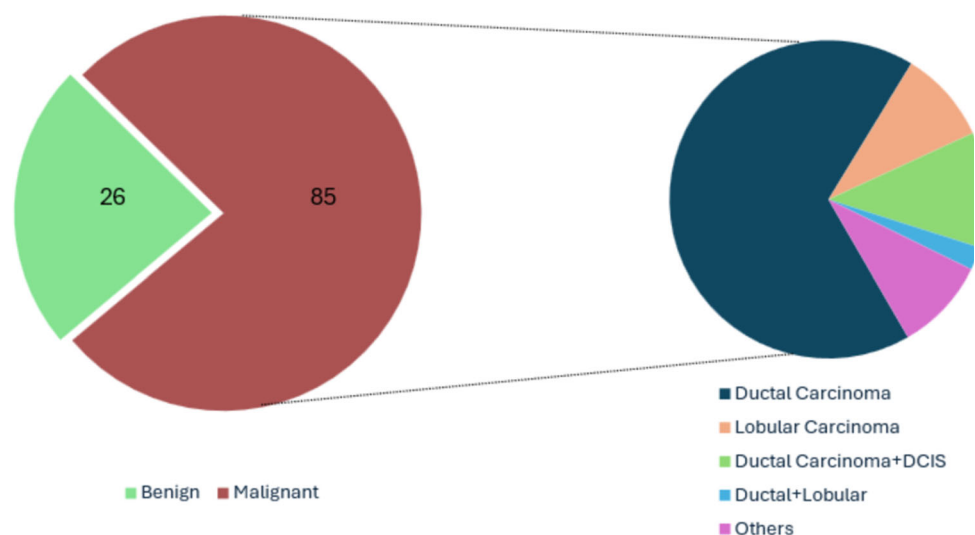
2.1 Dataset description

This work utilized the dataset from the online Her-2/neu scoring contest organized by the University of Warwick, the UK [37], along with the data provided by the pathologists. The Her-2 dataset consists of 172 whole slide images (WSIs) in NanoZoomer Digital Pathology (NDPI) format. These WSIs were obtained from 86 cases of invasive breast carcinoma. Each case includes an H&E-stained whole slide image and a DAB-stained whole slide image, making 172 -WSIs available for analysis. In this study, only the H&E-stained WSIs were considered. Among the 86 cases, one WSI was found to have a corrupted NDPI file. That case was not considered for further processing and analysis in the work. As all those Her-2 slides were malignant, we utilized 26 additional benign WSIs provided by two pathologists (R.M. and S.V.), retrieved from their anonymized consultation case archives. In total, we have 111 WSIs from those two different cohorts. Later, the malignant slides were again subdivided into different categories. From 85 malignant cases, 57 cases were reported as invasive ductal carcinoma (IDC), 10 cases were identified as invasive ductal carcinoma + DCIS (ductal carcinoma in situ), 8 cases were reported as invasive lobular carcinoma (ILC), 2 cases were reported as mixed invasive carcinomas (ductal and lobular), and the pathologists also identified 8 whole slide images of rare special types of breast cancers. In our study, we focused specifically on 57 cases of IDC and 8 cases of ILC. Other cancer subtypes were deliberately excluded from this research to maintain a concentrated analysis on these two predominant forms of breast cancer. Figure 1 shows the number of WSI for all the classes.

2.2 Data preparation

Initially, all WSIs were examined by two experienced pathologists (S.V. and R.M.) and categorized each as benign or malignant. The labels assigned by the pathologists were used as the ground truth for WSI-level scoring. Any disagreements between the pathologists were resolved through discussion until a unanimous decision was reached. Subsequently, the WSIs were divided into smaller patches of 1000×1000 pixels at 20x magnification. While most existing approaches generate patches at a 40x magnification level, our work aimed to investigate the performance of models on larger patches to enhance computational efficiency and better replicate the perspective of pathologists during their manual analysis. Each patch was then individually inspected by the pathologists using custom-built annotation software. Patches were labeled benign, malignant, or discarded if they were blurry or contained only very small regions with staining components. Patches identified as discarded were excluded from further experiments. A total of 28,428 patches were extracted, of which 14,214 were labeled as benign, 8019 as malignant, and 6195 were discarded. Due to the larger size of the benign WSIs, more benign patches were created. For the classification of ductal vs lobular cases, regions of interest (ROIs) were selected where cancer characteristics were most prominent, and patches were created from those regions of WSIs. The patch size was set at 3000×3000 pixels at 10x magnification, which was chosen to help the model better visualize and understand the pattern differences between ductal and lobular classes. The characteristics and differences were more prominent at the 10x magnification level compared to 20x. In total, 642 ductal patches were created from 57 ductal WSIs, and 444 lobular patches were created from 8 lobular WSIs.

Fig. 1 Visualization of dataset



Subject-wise cross-validation was conducted on all WSIs to generate two classification datasets. The first dataset consisted of patches labeled by pathologists as benign or malignant. The second dataset exclusively contained patches of ductal and lobular classes. Fivefold cross-validation was performed on both datasets to ensure reliable performance measurement of the models and the inclusion of patches from every WSI in the test set. It should be noted that there were fewer malignant patches compared to benign in the benign vs malignant classification, and fewer lobular cases compared to ductal in the ductal vs lobular classification. This imbalance could bias the model toward the class with more images, potentially leading to overfitting. To address this problem and introduce more variability in the training set, synthetic images were generated from the original images. Augmentation techniques including scaling, translation, rotation, and vertical and horizontal flipping were employed to create synthetic images. This helped to equalize the number of images in each class and create a more balanced training set [5].

2.3 Proposed method

This study introduces a novel and advanced framework for predicting cancer types using H&E-stained WSI files. The framework comprises three key stages: a cancerous patch classifier, a cancer subtype classifier, and a WSI-level classifier; additionally, the visualization technique Score-CAM (Score-Weighted Class Activation Mapping) is utilized to provide interpretability by highlighting critical regions that influence the model's decisions.

In the cancerous patch classifier stage, patches are generated from all WSIs and categorized as benign or malignant. Differentiating between benign and malignant patches is crucial for accurate cancer identification, and a deep learning model is trained specifically for this task. For the cancer subtype classifier stage, patches are classified into two categories: ductal carcinoma and lobular carcinoma. A separate model is trained to accurately classify these cancer subtypes. To predict the WSI-level result for both cancer and its subtypes, a majority voting technique with effective thresholds is employed. These thresholds were validated and approved by pathologists to ensure accuracy. Finally, Score-CAM visualizes the regions of an image that the model emphasizes in its predictions, providing valuable insights into the model's decision-making process and enhancing interpretability. The overview of the proposed method is shown in Fig. 2.

2.3.1 Cancerous patch classifier

The objective of this step is to accurately identify and differentiate between cancerous and non-cancerous tumor patches. In clinical terminology, a cancerous tumor is referred to as malignant, whereas a non-cancerous tumor is called benign [46]. This distinction is crucial for diagnosis and treatment planning. To achieve this, we classify each patch as either benign or malignant by analyzing its unique characteristics and features. Advanced image analysis techniques and machine learning models are employed to detect these features and make precise classifications.

The CNN model is composed of convolutional filters, which play a crucial role in identifying and extracting vital features from medical images. These filters operate by scanning the images and detecting patterns, edges, textures, and other significant details. Through this process, CNN can learn and understand the important characteristics within the images, thereby enhancing its ability to interpret and analyze medical data accurately and efficiently [42]. But recently Vision Transformers (ViTs) are becoming better than CNNs because they capture global context more effectively through self-attention mechanisms, which allows them to model long-range dependencies within images. They are also more flexible with varying input resolutions and can scale better with larger datasets and computational resources. Additionally, ViTs often exhibit improved performance in transfer learning scenarios, making them well-suited for a range of complex image analysis tasks [25, 30].

As previously mentioned, we created and annotated patches from the WSIs with the assistance of expert pathologists to ensure accurate labeling. We then split our data into training, testing, and validation sets, organizing them based on subjects to maintain consistency. Initially, our dataset was imbalanced, with an unequal distribution of benign and malignant images. To address this, we applied various augmentation techniques to the training set, including flipping, rotation, and zooming. These augmentations increased the diversity of the training data and balanced the number of images in each class, ensuring that both benign and malignant classes had an equal number of images. After preprocessing the dataset, we trained several deep learning models on the balanced training set. We evaluated the performance of these models and identified the top four best-performing ones. This selection included three models based on CNNs and one model based on a ViT network.

2.3.2 Cancer subtype classifier

Breast cancer exhibits a spectrum of subtypes, among which IDC and ILC are prevalent, comprising

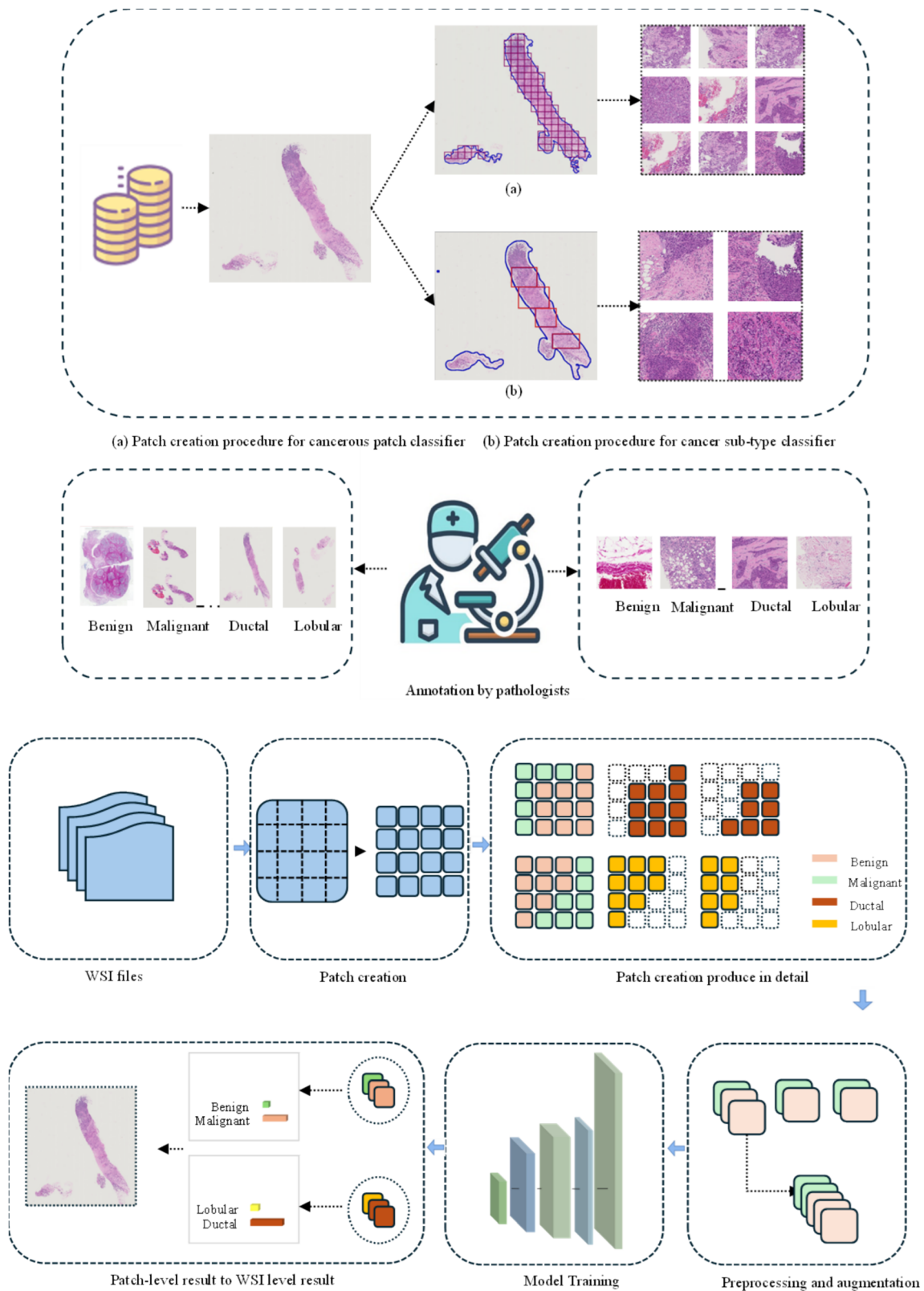


Fig. 2 Overview of the proposed method

approximately 70–80% and 5–15% of all invasive breast cancers, respectively [59]. During this phase, we utilized

deep learning models to differentiate between ductal and lobular cancer subtypes within the corresponding WSIs.

We extracted ROIs and created patches from there for ductal and lobular WSIs, emphasizing areas that prominently exhibited the characteristic features of each subtype. Following the preprocessing steps outlined in Sect. 2.3.1, we trained multiple deep learning models and evaluated their performance, identifying the top four performers, including three CNN models and one employing a ViT architecture.

2.3.2.1 DenseNet-201 DenseNet-201, a CNN architecture, has emerged as a formidable tool in the realm of deep learning, particularly in computer vision tasks. Proposed by Huang et al. in 2017, DenseNet-201 builds upon the concept of densely connected convolutional layers to enhance feature propagation and encourage feature reuse throughout the network [22]. Unlike traditional CNNs where each layer receives inputs only from the preceding layer, DenseNet-201 establishes direct connections between all layers within a dense block, thereby fostering richer feature representations. This characteristic not only facilitates gradient flow during training but also mitigates the vanishing-gradient problem, enabling more effective training of deeper networks. With its remarkable performance on benchmark datasets such as ImageNet, DenseNet-201 has become a cornerstone in various applications including image classification, object detection, and segmentation. Moreover, its compact yet expressive architecture makes it well-suited for deployment in resource-constrained

environments. The architecture of DenseNet-201 is shown in Fig. 3.

2.3.2.2 MobileNet architecture (MobileNetV2)

MobileNetV2 is a CNN architecture designed for efficient image classification, particularly suited for mobile and edge devices [21]. Introduced by Sandler et al. in 2018, MobileNetV2 builds upon the original MobileNet architecture with several key enhancements that improve performance and efficiency [41]. The core innovation of MobileNetV2 is the use of inverted residuals and linear bottlenecks. This design involves using a thin bottleneck layer and expanding it to a higher-dimensional space where depth-wise convolutions are applied. The expansion and projection operations are performed using 1×1 convolutions, which help preserve the model's expressiveness while maintaining efficiency by reducing the number of parameters and computational complexity. Like its predecessor, MobileNetV2 utilizes depth-wise separable convolutions. This type of convolution factorizes a standard convolution into a depth-wise convolution and a pointwise convolution (1×1 convolution), significantly reducing the computational cost and the number of parameters while maintaining high performance. The architecture is particularly noted for its efficient layer design, which enhances both speed and accuracy, making it suitable for deployment in resource-constrained environments such as mobile

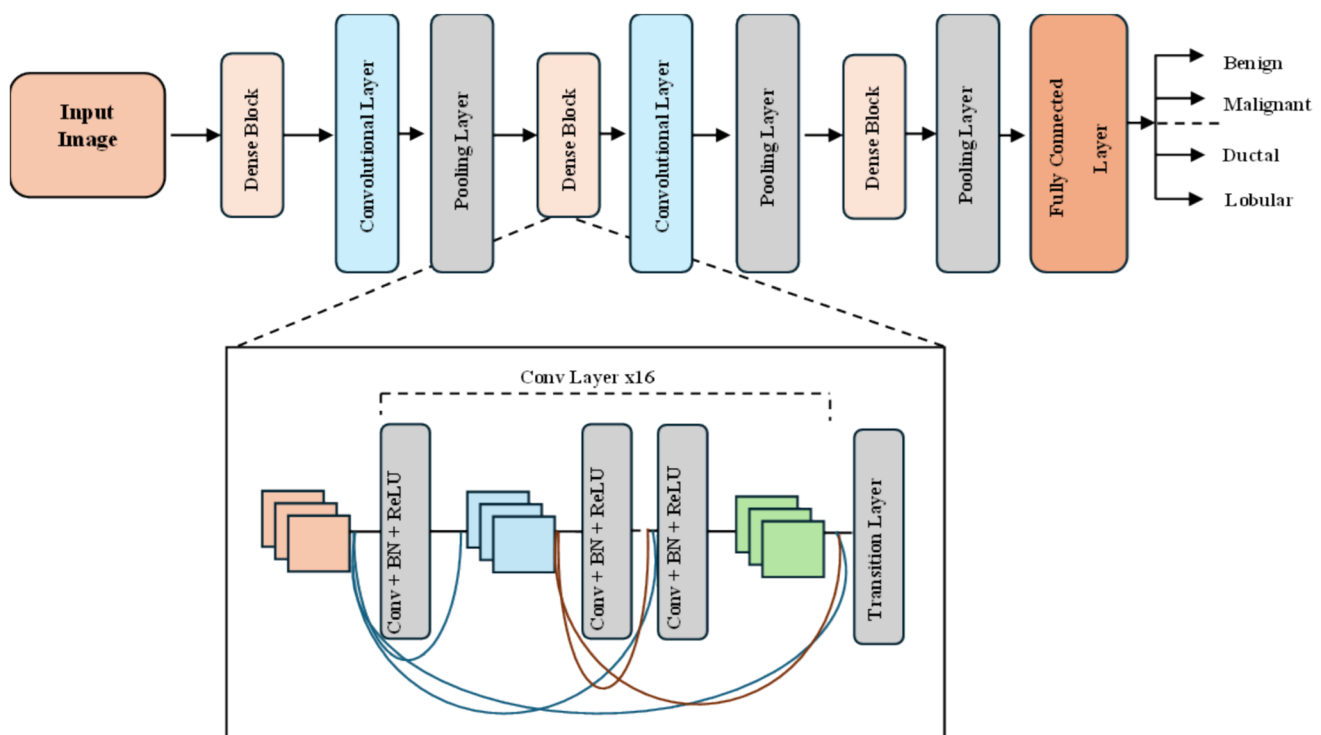


Fig. 3 Architecture of DenseNet-201 [22]

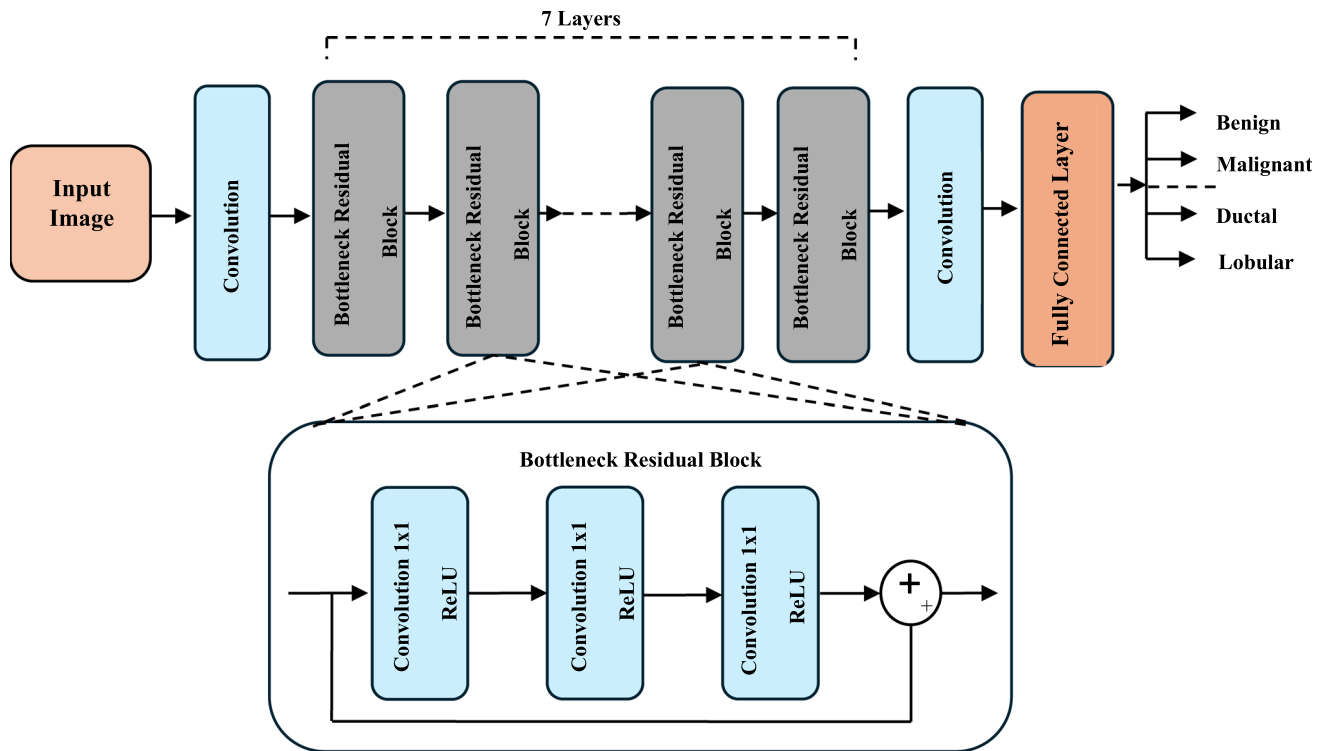


Fig. 4 Architecture of MobileNetV2 [21]

devices. In Fig. 4, the architecture of MobileNetV2 is presented.

2.3.2.3 Ensemble of DenseNet-201 and MobileNetV2 The Ensemble model is a sophisticated ensemble architecture that integrates components from DenseNet-201 and MobileNetV2 to enhance feature extraction and classification capabilities [22, 41]. Specifically, it utilizes the initial layers from DenseNet-201 and MobileNetV2, capturing diverse feature representations from both models. This integration is achieved by creating three distinct sets of blocks from each model: the first seven, the first nine, and all layers, which are then concatenated to form comprehensive feature maps. The model includes two auxiliary classifiers and a final classifier. The auxiliary classifiers process intermediate feature maps using average pooling, convolutional layers, batch normalization, ReLU activation, and adaptive average pooling, followed by a fully connected layer and log softmax for classification. The final classifier processes the complete concatenated feature map, similarly, employing adaptive average pooling and a fully connected layer for the final classification. This design allows the ensemble model to leverage the complementary strengths of DenseNet's dense connectivity and MobileNet's efficient depth-wise separable convolutions, enhancing its overall performance on complex

classification tasks. Figure 5 illustrates the architecture of the ensemble model.

2.3.2.4 Vision Transformer (ViT) ViT models have emerged as a powerful approach in the realm of computer vision, leveraging principles from transformer architectures initially designed for natural language processing [13]. These models provide unique advantages, especially in medical imaging, due to their ability to capture and process complex patterns in visual data [9]. The architecture of ViT begins with the input image being divided into a grid of fixed-size patches, typically 16×16 pixels. Each of these patches is then flattened into a vector and linearly projected to a lower-dimensional space, resulting in what is known as patch embeddings. This step transforms the image into a sequence of patch embeddings, much like how words in a sentence are treated in language models [53]. To retain spatial information, which is crucial for understanding the structure within images, position embeddings are added to the patch embeddings. These position embeddings encode the spatial location of each patch within the original image, ensuring that the model can maintain an understanding of the image's layout and structure. Once the patches are embedded and positional information is added, the sequence of patches is fed into a standard transformer encoder. This encoder comprises multiple layers of multi-head self-attention mechanisms and feed-forward neural

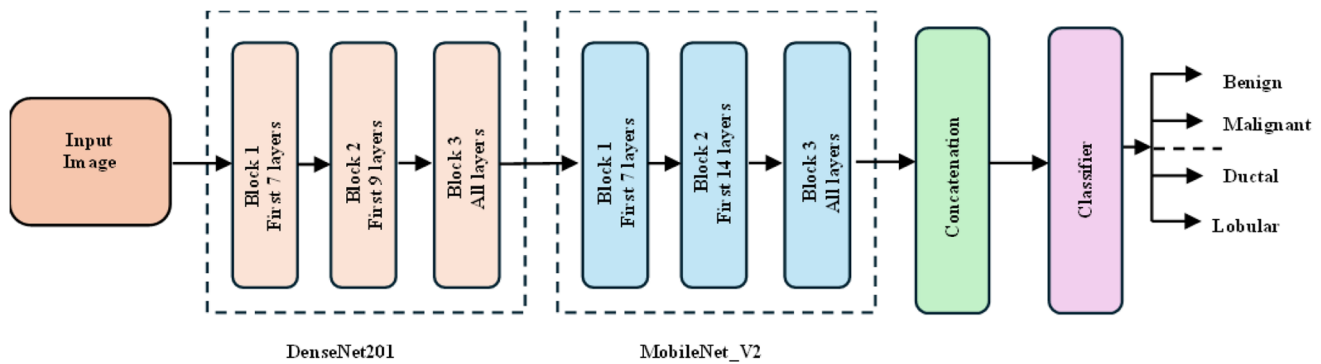


Fig. 5 Architecture of ensemble model

networks. The self-attention mechanism enables the model to weigh the importance of each patch in the context of the entire image, allowing it to focus on relevant features and patterns [53].

In the context of medical images, Vision Transformers offer several significant advantages. Firstly, their ability to model long-range dependencies and capture global context is particularly beneficial [4]. Medical images, such as MRI scans or histopathology slides, often contain subtle anomalies spread across different regions. ViTs can integrate information across the entire image, making them adept at identifying these anomalies. Moreover, ViTs are inherently more flexible than CNNs when dealing with varying image resolutions and aspect ratios [36] this flexibility is valuable in medical images, where images can vary widely in terms of resolution and scale. The patch-based approach of ViTs allows them to process images at different scales more effectively. Another advantage of ViTs is their ability to learn from large-scale pretraining on diverse datasets, which can then be fine-tuned on specific medical images tasks [52]. This transfer learning capability allows them to leverage knowledge from general visual features and apply it to specialized medical domains, improving performance even with limited labeled medical data. In summary, ViT models represent a significant leap forward in medical imaging due to their ability to capture complex patterns and global context within images [4, 13]. Their flexible architecture and capacity for effective transfer learning make them well-suited for the nuanced and varied challenges presented by medical image analysis [36, 52]. The architecture of the vision transformer is demonstrated in Fig. 6.

2.3.3 WSI-level classifier

2.3.3.1 Cancer classifier (malignant or benign) To determine the overall classification for a patient's WSI, we applied a majority voting strategy based on the predictions of individual patches [14]. Each patch from the WSI was

independently classified by the model as benign or malignant. The final WSI-level diagnosis was then derived by aggregating these patch-level predictions. Specifically, if more than 99% of the patches were predicted to be benign, the patient was classified as having a benign condition. Conversely, if less than 1% of the patches were predicted to be malignant, the patient was classified as having cancer. This stringent criterion ensures high sensitivity, prioritizing the identification of malignant cases to minimize the risk of missing a cancer diagnosis.

2.3.3.2 Subtype of cancer For cancerous cases, a more detailed classification into ductal or lobular carcinoma was carried out based on patch-level predictions. This secondary classification step involved analyzing the malignant parts of the ductal and lobular to determine the predominant subtype. If more than 50% of the patches were predicted to be ductal, the patient was diagnosed with ductal carcinoma. Conversely, if the majority of patches were predicted to be lobular, the patient was diagnosed with lobular carcinoma. Ductal and lobular carcinomas can exhibit different behaviors and may respond differently to treatment, making this distinction significant in clinical practice. This approach allows for nuanced diagnosis by not only identifying the presence of cancer but also specifying its subtype, which is crucial for treatment planning and prognosis.

2.3.4 Visualization of cancerous patch

Score-weighted class activation mapping (Score-CAM) is an innovative approach in the field of deep learning that enhances the interpretability of CNNs [55]. Score-CAM is a variant of class activation mapping (CAM) techniques, designed to visualize the regions of an image that contribute most to CNN's classification decision. Unlike traditional CAM methods that rely on the gradients of the target class score concerning the activation maps, Score-CAM uses the class scores directly to weigh the activation

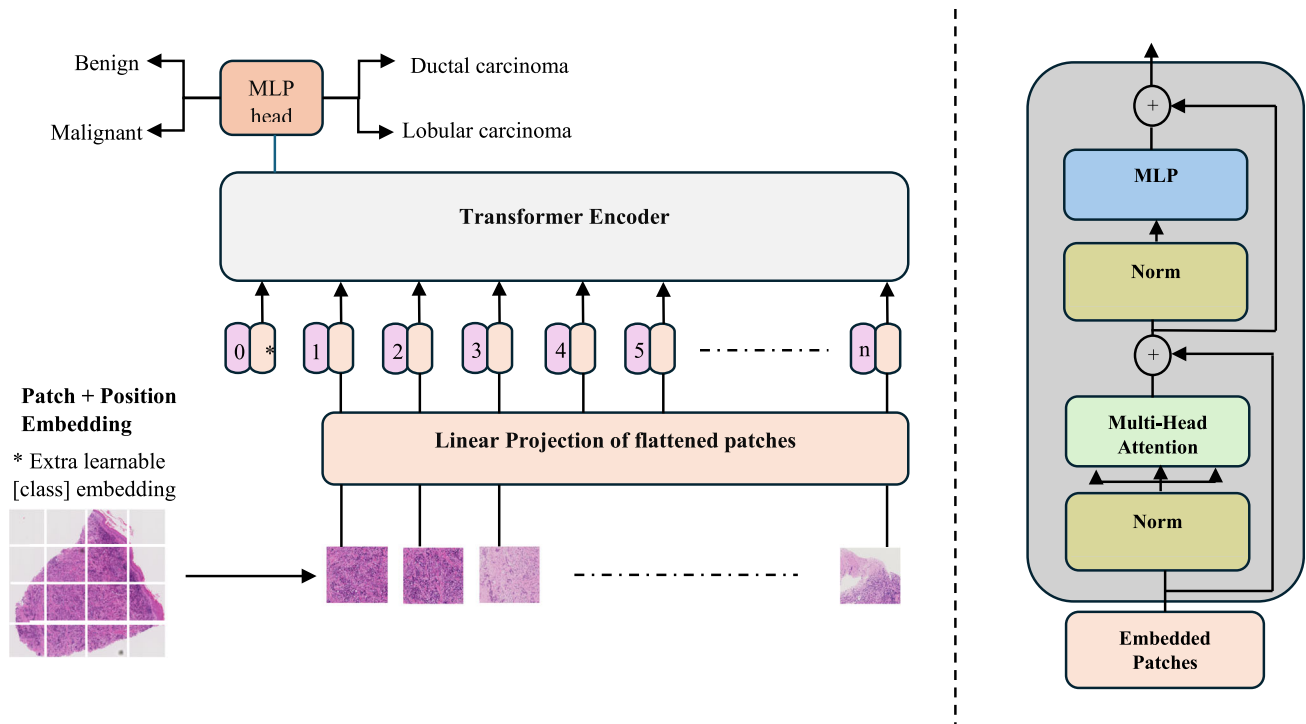


Fig. 6 Architecture of vision transformer [13]

maps. This approach eliminates the need for gradients, making it more robust and less susceptible to noise [55]. In the context of cancer classification using histopathology images, Score-CAM plays a critical role by providing visual explanations of the model's decision-making process, thereby aiding in more accurate and transparent diagnoses [43]. Deep learning models, particularly CNNs and ViTs, offer high accuracy and efficiency in automation, but their interpretability remains a key challenge [23]. By visualizing the regions the model focuses on, pathologists can better understand and trust its predictions, ensuring reliable and interpretable decisions. This transparency helps researchers enhance training data to cover more relevant patterns, leading to more robust models.

In our proposed system, we explored various visualization techniques for interpretability. Ultimately, we selected Score-CAM for its superior clarity and insight in visualizing the model's decision-making process, using it to generate activation maps for the test images after training. We provided visualizations for cancerous patches and their subtypes, such as ductal and lobular carcinomas. These activation maps highlight the regions of the histopathology images that the model deems most important for classification. By applying Score-CAM, we produced heatmaps that visually represent these critical regions, enhancing the interpretability of the model's decisions. These heatmaps enable pathologists to verify the model's focus areas, ensuring that the highlighted regions

correspond to medically relevant structures. This process not only aids error analysis and model refinement but also builds trust in the artificial intelligence system's diagnostic capabilities.

2.4 Training and testing methodology

The proposed architecture's initial classification stages involve deep learning networks designed for patch classification. For cancerous patch classification, we conducted a subject-wise data split based on fivefold cross-validation, ensuring that 80% of the data was allocated to the training set and 20% to the test set. Additionally, we reserved 10% of the training data for validation purposes. The subject-wise split ensures that all patches from a specific case are placed consistently within the same set—either in training, testing, or validation—to prevent data leakage issues and provide a more accurate evaluation of the model's performance. Data leakage occurs when information from the test set inadvertently influences the training process, leading to overly optimistic performance estimates. By using a subject-wise split, we ensure that the model is evaluated on entirely unseen data, thus providing a more accurate assessment of its generalization capability [6]. After this split, the training set comprised patches from 80 WSIs, the test set included patches from 22 WSIs, and the validation set contained patches from 9 WSIs. To generate patches from the WSIs, we extracted all possible patches with

dimensions of 1000 by 1000 pixels at a 20x magnification level.

For subtype cancer patch classification, we applied the same subject-wise splitting strategy to the WSIs of ductal and lobular carcinoma. This resulted in 41 WSIs of ductal carcinoma and 5 WSIs of lobular carcinoma in the training set. The remaining WSIs were split into test and validation sets. For this classification task, we selected specific ROIs and created patches of size 3000 by 3000 pixels at a 10x magnification level, which proved optimal for distinguishing between ductal and lobular subtypes. These ROIs were carefully chosen to capture the most informative areas of the tissue, highlighting the distinct histological features necessary for accurate subtype classification. This approach ensures that the model can effectively learn and differentiate between the nuanced patterns present in ductal and lobular carcinoma, leading to more precise and reliable subtype predictions.

For patient-level cancer prediction, we implemented a majority voting mechanism with a stringent threshold of less than 1%. This means that if even less than 1% of the patches indicate malignancy, the entire WSI is classified as malignant. This threshold ensures the robustness of our model by providing high sensitivity in detecting cancer. It prioritizes identifying even the smallest signs of malignancy, which is crucial for early diagnosis and treatment. By adopting such a conservative threshold, we minimize the risk of missing any malignant regions, ensuring that the model errs on the side of caution and maximizes patient safety. In contrast, for WSI-level subtype prediction, we used a 50% threshold. Given that both ductal and lobular carcinoma are forms of breast cancer and equally hazardous, a balanced 50% threshold was deemed most appropriate. This approach ensures that if more than 50% of the patches indicate a particular subtype, the entire WSI is classified accordingly. This balanced threshold ensures the model accurately classifies both ductal and lobular carcinomas, which is critical for determining appropriate treatment plans and managing patient outcomes effectively.

In this study, models pre-trained on ImageNet were utilized, resizing all patch images to 224×224 pixels to match the required dimensions. The images were normalized using the standard distribution. The training procedure for the deep learning models consisted of 150 epochs with a batch size of 16 and an early stopping patience of 10 epochs, saving the best model based on validation performance. We experimented with various optimizers, but the Adam optimizer yielded the best results due to its adaptive learning rate and was used throughout the training process. Preliminary training involved experimenting with various learning rates, and a learning rate of 0.0001 was chosen for its superior performance.

In our experimental setup, we utilized two graphical processing units (GPUs): an NVIDIA GeForce RTX 4090 and an AMD Radeon™ graphics card. The system was equipped with 128 GB of RAM and powered by an AMD Ryzen 9 7950X3D 16-core processor. All training and experimentation were conducted using Visual Studio Code.

3 Results

3.1 Evaluation matrices

To assess the performance of various classifiers, we employed multiple evaluation metrics, which are derived from the confusion matrix. Specifically, the confusion matrix provides key values—True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN)—that allow for the calculation of metrics such as Accuracy (A), Recall/Sensitivity (R), Precision (P), Specificity (S), and F1-Score (F1) [48]. The mathematical formulations for the five-evaluation metrics are depicted in Eqs. (1–5). The confusion matrix serves as the foundation for calculating these metrics, offering a comprehensive view of the model's performance across different classes. Each metric was calculated separately for each class to provide a detailed understanding of the classifier's performance across different categories. For an overall performance measure, we calculated the weighted average of each metric, with weights proportional to the class distribution. This approach provides a balanced measure of the model's effectiveness across both classes without being skewed by class imbalance. The weighted average for each metric is calculated using Eq. (6).

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$P = \frac{TP}{TP + FP} \quad (2)$$

$$R = \frac{TP}{TP + FN} \quad (3)$$

$$F1 = 2 \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (4)$$

$$S = \frac{TN}{TN + FP} \quad (5)$$

$$\text{Weighted Average Metric} = \sum_{i=1}^n \frac{N_i}{N} \times \text{Metric}_i \quad (6)$$

In this equation, n is the number of classes (for instance, 2 in cases of binary classification), N_i is the number of instances in class i , N is the total number of instances across all classes, and Metric_i is the metric value (such as precision or recall) for class i . For example, to calculate

weighted average precision, this formula is applied using precision values for each class, weighted by their dataset proportions.

Metrics were calculated from a fivefold cross-validation, with predictions concatenated for final analysis. This approach ensures metrics reflect the model's performance across the entire dataset, providing a robust evaluation across classes and folds.

3.2 Cancerous patch classifier performance

In this section, we present the performance metrics of the deep learning models trained for the identification of cancerous patches.

DenseNet-201 achieved an accuracy of 94.5%, with a recall of 94.5%, an F1-score of 94.56%, and a specificity of 96.48%. MobileNetV2 outperformed DenseNet-201 in terms of accuracy, precision, sensitivity, and F1-score, demonstrating a precision of 96.09%, recall of 96.1%, and an F1-score of 96.09%. Notably, the specificity of DenseNet-201 was higher than that of MobileNetV2, indicating superior performance in identifying malignant classes. So, to enhance the classifier's performance, we implemented an ensemble model that combines DenseNet-201 and MobileNetV2. This ensemble model surpassed the individual performances of DenseNet-201 and MobileNetV2, achieving accuracy, precision, sensitivity, and an F1-score of 96.59%. The Vision Transformer model exhibited superior performance over all previously evaluated models, attributed to its enhanced capability to capture global context more effectively than CNNs. The ViT Model achieved the highest accuracy among the evaluated models, with a score of 96.74%, and demonstrated equal precision, sensitivity, and F1-score values, all at 96.74%. These metrics indicate that the Vision Transformer model provided the best overall performance.

Given that accuracy alone does not comprehensively reflect the models' performance, we include the confusion matrices for a more detailed evaluation [17]. Figure 7 presents the confusion matrix of the cancerous patch classifier. The confusion matrices reveal that the Vision Transformer model excels in correctly identifying both malignant and benign classes. Although DenseNet-201 is effective in identifying malignant classes, it also misclassifies a significant number of benign cases as malignant, which is not desirable for cancer detection. In summary, the Vision Transformer model is identified as the most effective model for cancerous patch classification, considering its superior performance metrics and ability to accurately distinguish between malignant and benign patches.

3.3 Cancer subtype classifier performance

In this section, we discuss the performance of the deep learning models trained to classify cancer subtypes. For subtype classification, Ensemble of DenseNet-201 and MobileNetV2 gained a precision of 85% and an accuracy and recall of 84.07%, which was outperformed by DenseNet-201 with an accuracy of 86%, with a precision of 87.13% and a recall of 86.01%. Similar to its performance in cancer prediction, DenseNet-201 performs well in identifying the ductal class but struggles with the lobular class. MobileNetV2 gained better results than the previously mentioned models in classifying cancer subtypes. It achieved a precision of 88.6%, accuracy of 88.12%, recall of 88.12%, and an F1 score of 87.9%, indicating superior performance compared to the previously mentioned models.

The ViT model exhibited the best performance among all models in classifying cancer subtypes. It achieved a precision of 90.3%, accuracy and sensitivity of 89.78%, an F1-score of 89.6%, and specificity of 86.54%. These results indicate that the ViT model outperformed all other models in all evaluated metrics. Figure 8 demonstrates the confusion matrix for the cancer subtype classifier. The confusion matrices further illustrate the superior performance of the ViT model, showing the least misclassification rates for both ductal and lobular classes. However, it is evident from the confusion matrices that the misclassification rate for the lobular class was higher than for the ductal class. This discrepancy is likely due to the limited data available for the lobular class, with only 8 cases, resulting in less variability and impacting the models' performance. Increasing the number of lobular subtype cases would likely improve the performance of all models. In summary, the ViT model is the most effective model for classifying cancer subtypes, demonstrating the highest performance metrics and the lowest misclassification rates.

3.4 WSI-level classifier performance

In the final stage of our framework, we aimed to predict the whole slide-level results for cancer and its subtypes using a majority voting technique. By applying majority voting to the patch-based results of the Vision Transformer model in cancer classification, we achieved an accuracy and sensitivity of 98.19%, along with an F1-score of 99.06%, in WSI-level prediction. The confusion matrix demonstrates that there were no misclassifications in identifying the malignant class. This outcome supports our threshold criterion: If less than 1% of the patches suggest cancer, the WSI is classified as cancerous. The absence of

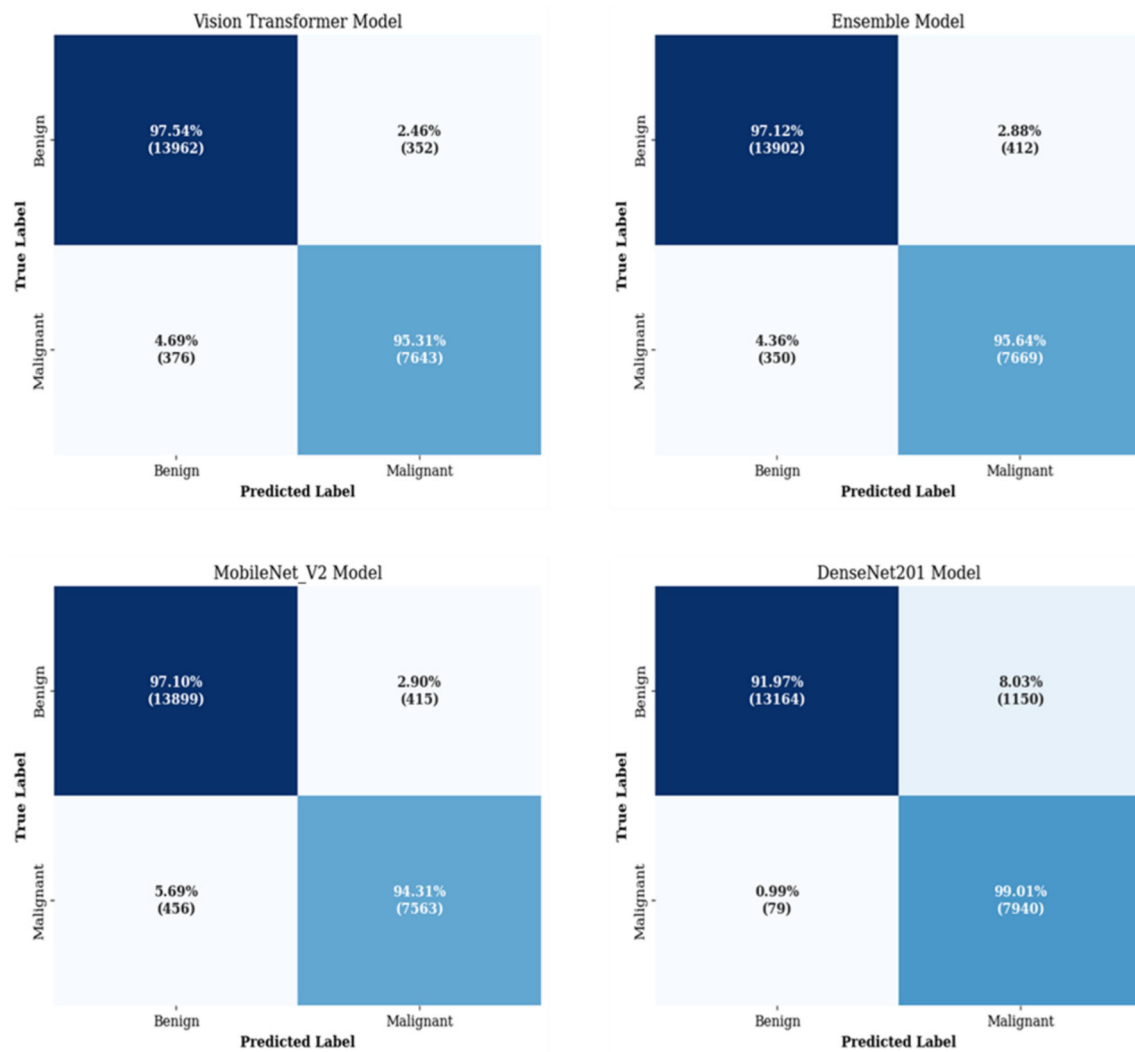


Fig. 7 Confusion matrix of cancerous patch classifier

misclassification in the malignant class further validates this threshold.

For the classification of cancer subtypes at the patient level, we employed a similar methodology. Using a 50% threshold on the patch-based results for subtype classification, our model achieved an accuracy of 95.38%, a sensitivity of 95.38%, a precision of 96.89%, an F1-score of 96.13%, and a specificity of 96.73%. The confusion matrix reveals that our model accurately predicted 55 WSI labels for the ductal class and 7 WSI labels for the lobular class. Figure 9 represents the confusion matrix of the WSI-based classifier. Despite the limitations of the dataset, the results are promising for future applications in medical diagnostics.

Table 1 demonstrates the results for all three types of classifiers.

3.5 Visualization of cancerous patch

In our proposed system, as part of the validation process, we employed Score-CAM post-training to generate activation maps for the test images. We created visualizations for cancerous regions and their subtypes, including ductal and lobular carcinomas. These activation maps emphasize the critical areas of the histopathology images for classification, with Score-CAM producing heatmaps to visually highlight these key regions. Figure 10 displays the heatmap visualization of a cancerous patch classifier, showcasing four malignant cases from the test set. Figure 11 illustrates the heatmap visualization from the cancer subtype classifier, featuring two ductal and two lobular cases from the test set.

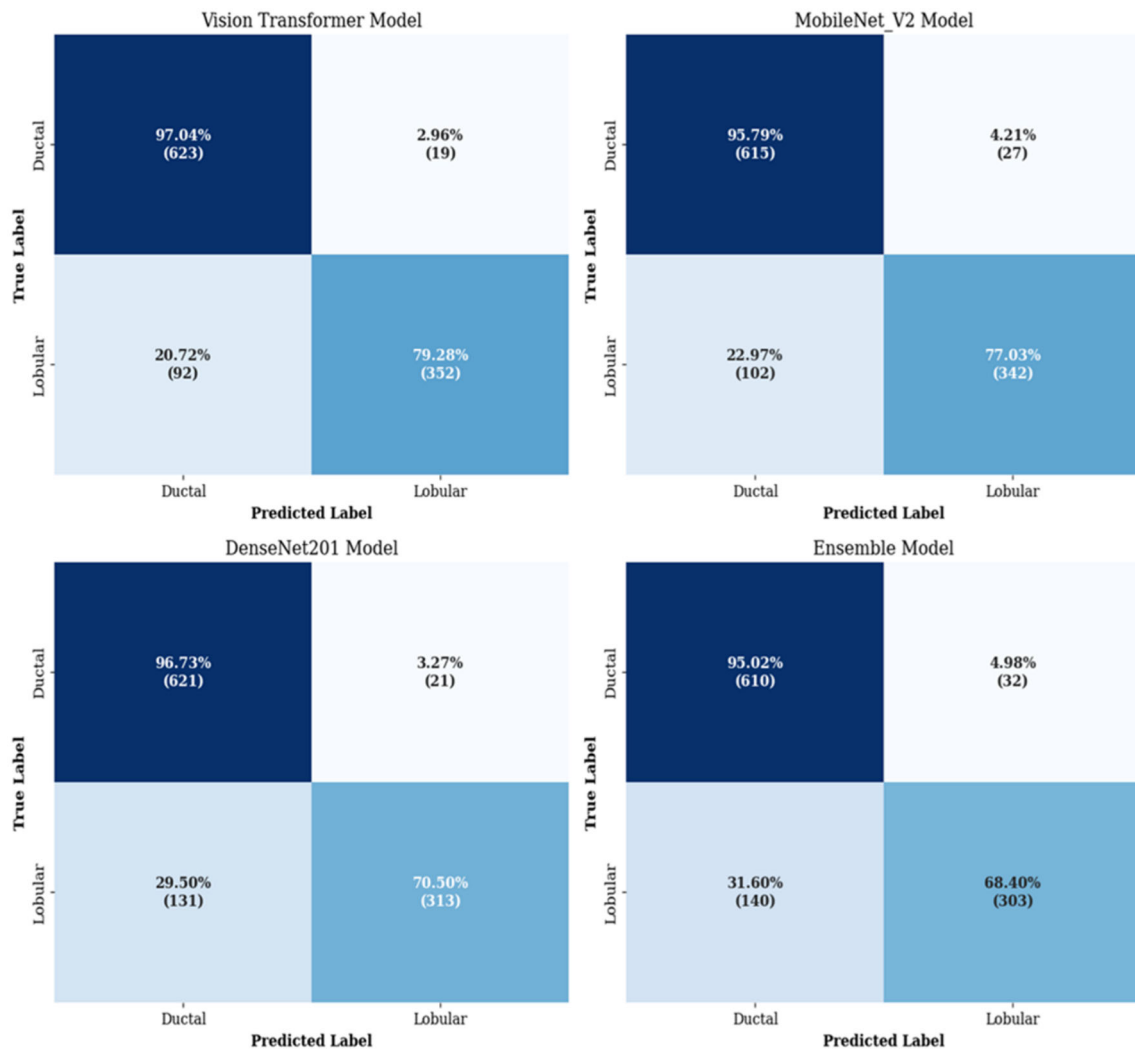


Fig. 8 Confusion matrix of cancer subtype classifier

4 Discussion

In this study, we have presented a novel framework for the detection of breast cancer. The methods include the detection of breast cancer and its subtypes at both the image level and patient level with the utilization of a large dataset, along with visualization using Score-CAM to effectively highlight the areas that are most affected or relevant for the prediction.

Obayya et al. [34] proposed an arithmetic optimization algorithm with a deep learning-based histopathological breast cancer classification (AOADL-HBCC) technique. For the dataset, they used BreakHis's 200x and 100x magnification level images. They proposed a deep belief network (DBN) model, which achieved an accuracy of 96%. Their classification is limited to the patch level, only addressing the binary classification of benign versus malignant. Srikantamurthy et al. [49] developed a hybrid

model using a CNN and a long short-term memory recurrent neural network (LSTM RNN) to classify breast histopathology images at both binary and multiclass levels. Their classification was done at the patch level. They utilized the BreakHis dataset, which consists of 2480 benign patches and 5429 malignant patches across four different magnification levels (40x, 100x, 200x, and 400x). This means for each magnification level; they had a smaller number of patches compared to our study. For binary classification, they achieved an accuracy of around 99% at three magnification levels (40x, 100x, and 200x) and 98% at the 400x magnification level. For multiclass classification, their accuracy ranged from 88 to 96% across all four magnification levels. It is noteworthy that their classification was confined to patch levels. This approach falls short in terms of interpretability and fails to provide a comprehensive evaluation of the model's performance, raising concerns about its practical applicability and robustness.

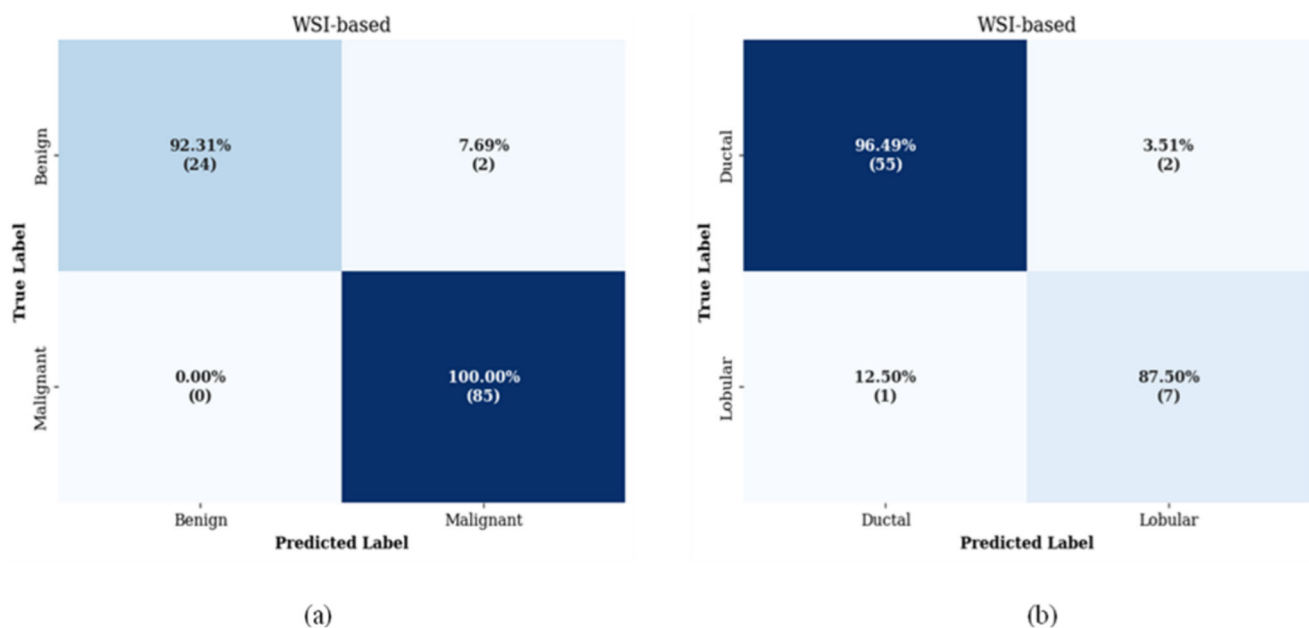


Fig. 9 **a** Confusion matrix of WSI-level cancer classification. **b** Confusion matrix of WSI-level cancer subtype classification

Table 1 Overview of evaluation metrics

	Models	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Specificity (%)
Cancerous patch classifier	DenseNet-201	94.5	95.07	94.5	94.56	96.48
	MobileNet_V2	96.1	96.09	96.1	96.09	95.31
	Ensemble	96.59	96.59	96.59	96.59	96.17
	Vision Transformer	96.74	96.74	96.74	96.74	96.11
Cancer subtype classifier	Ensemble	84.07	85	84.07	83.58	79.19
	DenseNet-201	86	87.13	86.01	85.57	81.22
	MobileNet_V2	88.12	88.6	88.12	87.9	84.7
	Vision Transformer	89.78	90.3	89.78	89.6	86.54
WSI-based classifier	Cancerous	98.16	100	98.16	99.04	100
	Cancer Subtype	95.38	96.89	95.38	96.13	96.73

Zhou et al. [60] employed a resolution adaptive network for both binary and multiclass classification, utilizing two publicly available datasets: BreakHis and BATCH. For the BATCH dataset, they achieved a patch-level accuracy of 97% for binary classification and 96% for multiclass classification. Notably, both datasets were smaller than the ones used in our study. The BreakHis dataset contains images at four different magnification levels, and the computational cost increases with higher magnification. Zhou et al. mentioned two techniques for aggregating decisions from patch level to image level, but they did not detail their methodology. Roy et al. [39] focused on a patch-based system for breast histopathology image classification using the BATCH 2018 dataset, which comprises only 400 H&E-stained patches. They proposed a

classification system operating in two modes: one patch one decision (OPOD) and all patches one decision (APOD). The OPOD mode achieved a patch-wise classification accuracy of 77.4% for multiclass classification and 84.7% for binary classification. In contrast, the APOD technique attained an image-wise classification accuracy of 90% for four classes and 92.5% for binary classification on the split test set.

Das et al. [10] proposed a multiple-instance learning (MIL)-based deep convolutional neural network (CNN) architecture, utilizing three datasets: BreakHis (for 40x and 200x magnification levels), IUHPL, and UCSB. For binary patch-level classification, their model achieved accuracies of 93.06, 96.63, and 95.83% for the respective datasets. In their approach, each slide is represented as a bag of

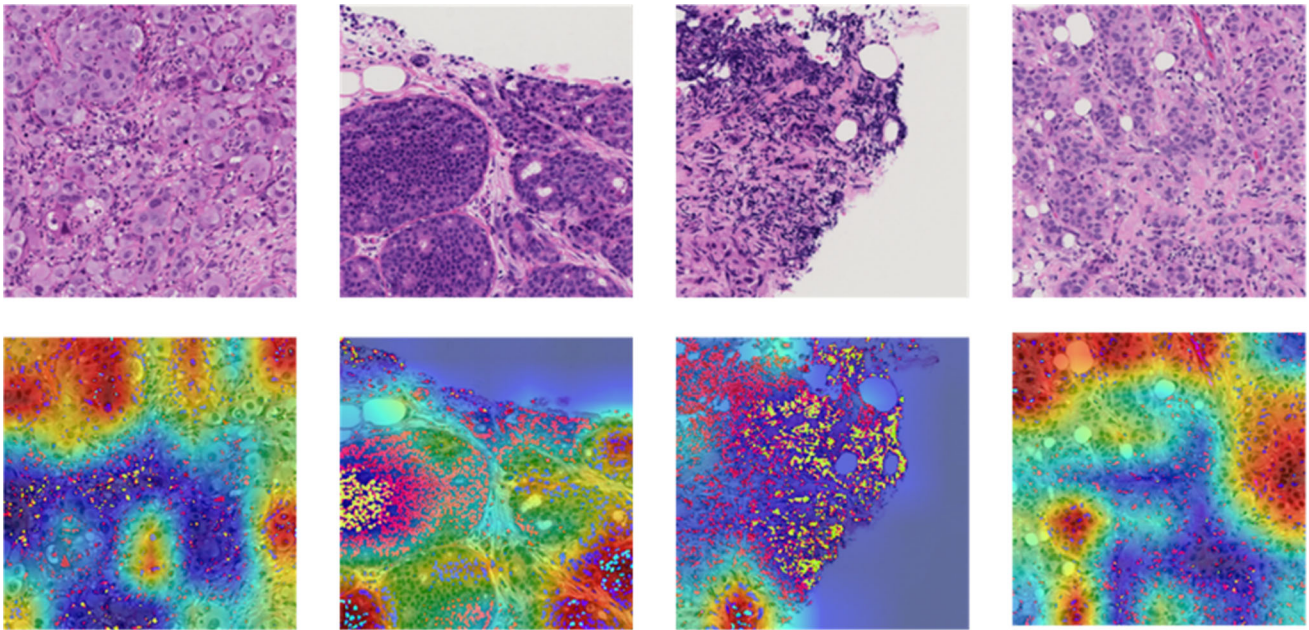
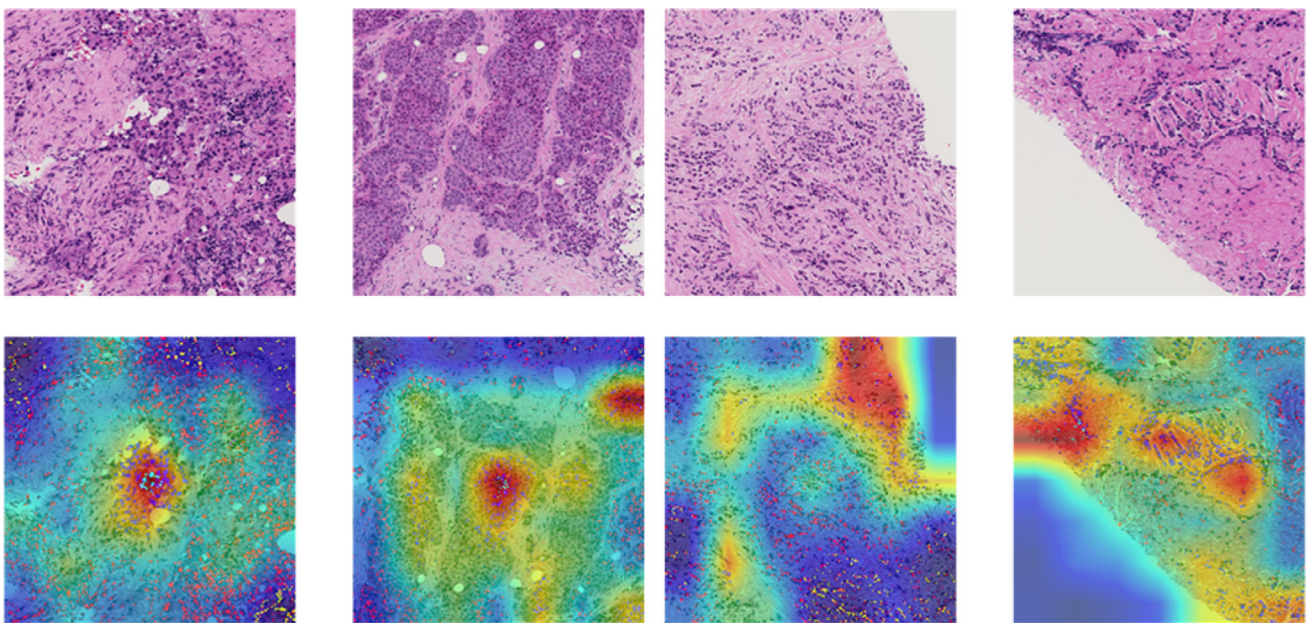


Fig. 10 Visualization for cancerous patch classifier



(a) Visualization for ductal carcinoma

(b) Visualization for lobular carcinoma

Fig. 11 Visualization for cancer subtype classifier

extracted patches, and only the bag label is used for training. A WSI is labeled benign if all its patches are benign and labeled malignant even if a single patch contains malignant cells. However, this method can create ambiguity, as WSIs can contain both benign and malignant cells. Identifying all patches in this manner may lead to misclassification of benign patches as malignant. In contrast, our study took a different approach. All generated

patches were examined by expert pathologists and labeled as either benign or malignant. This ensures that the model only sees malignant patches in the malignant class, without any benign patches, regardless of the overall WSI diagnosis. This methodology eliminates ambiguity and enhances the accuracy of patch-level classification by providing clearer distinctions between benign and malignant patches.

Table 2 Comparison with other existing approaches of breast cancer classification

References	Dataset	Approach	Result
[34]	BreakHis (40x and 100x) (82 samples, 4076 patches)	Deep belief network (DBN)	Accuracy of 96% for patch-level binary classification (benign vs malignant)
[49]	BreakHis (82 samples, 7909 patches)	hybrid model using CNN and LSTM	For binary classification accuracy of 98% for 400x and multiclass classification accuracy of 92.51% for 400x, both at patch level
[60]	BreakHis and BATCH 2018 (92 samples, 8309 patches)	Resolution Adaptive Network	Patch-level accuracy of 97% for binary classification and 96% for multiclass classification Image level accuracy of Binary classification is 98% and Multiclass classification accuracy is 97%
[39]	BATCH 2018 (10 samples, 400 patches)	Optimal architecture of CNN	Patch level: 77.4% for multiclass classification and 84.7% for binary classification, Patient level: 90% for four-class and 92.5% for binary classification
[10]	BreakHis (40x and 200x), IUHPL, and UCSB (180 samples, patches not mentioned)	Multiple-Instance Learning (MIL)-based Deep CNN architecture	Patch-level: binary classification, accuracy of 93.06%, 96.63%, and 95.83% for the respective datasets
Proposed approach	Dataset from two different sources (University of Warwick + provided by pathologists): A total of 111 WSIs consisting of 28,428 patches	Deep learning models (DenseNet-201, MobileNetV2, Ensemble of DenseNet-201 and MobileNetV2, Vision Transformer)	Cancerous patch classifier: Accuracy: 96.74% Cancer subtype classifier: Accuracy: 89.78% WSI-level classifier: Cancerous: Accuracy of 98.16% Cancer Subtype: Accuracy of 95.38%

In reviewing the current state-of-the-art research in histopathological image classification, it is evident that most prior studies have predominantly utilized BreakHis dataset. This dataset comprises images at four different magnification levels: 40x, 100x, 200x, and 400x. While these high magnification levels provide detailed images, they also come with significantly higher computational costs. Our study aimed to address this issue by reducing the magnification levels used in image analysis. For binary classification of cancerous patches, we utilized images at a 20x magnification level. For subtype classification, we further reduced the magnification to 10x. Additionally, our dataset included a total of 22,233 patches for binary classification, surpassing the size of any existing dataset in this domain. Another key innovation in our work is the provision of both patch-level and WSI-level classification techniques, encompassing both cancer detection and subtype classification. Our proposed model achieved an impressive accuracy of 96.74% for cancerous patch classification and 98.19% for WSI-level cancer prediction. For

the classification of cancer subtypes at the patch level, our model achieved an accuracy of 89.78%, while the WSI-level subtype classification achieved a precision of 96.8%. Furthermore, the visualization of the Score-CAM for cancerous cells reinforced the reliability of our proposed model.

Before comparing with existing literature, it is important to note that our dataset and experimental conditions are uniquely tailored to this study, and direct replication in prior work is absent. Therefore, while we provide these comparisons for context, differences in dataset composition, sample size, and methodological details should be considered when interpreting the results. Table 2 provides a comparison between our approach and other existing methods.

Despite our significant results, there is still room for improvement in the classification of cancer subtypes. The model's performance was hindered by the limited variability in the dataset, particularly the scarcity of lobular carcinoma data compared to ductal carcinoma data, leading

to higher misclassification rates for this subtype. Increasing the dataset size for lobular carcinoma could mitigate this issue. Moreover, the integration of multimodal datasets that include both image data and clinical information could enhance model performance. Future work will focus on leveraging deep learning to develop a comprehensive software system that offers customizable training and classification options specified by the user. This approach will enhance the potential for customization and adaptability, enabling more effective and scenario-specific applications in histopathological image classification.

5 Conclusion

This research introduces a novel advanced framework for predicting breast cancer and its subtypes using H&E-stained WSI files. The framework is composed of three critical stages: a cancerous patch classifier, a cancer subtype classifier, and a WSI-level classifier. Additionally, our study features the visualization of Score-CAM, which enhances interpretability by highlighting the regions influencing the model's predictions, aids validation by aligning with pathologists' identified features, facilitates error analysis by revealing misleading regions, and builds trust by providing transparent decision-making insights. All WSIs in the dataset were meticulously annotated at both the WSI and patch levels by two expert pathologists. To accurately identify patches containing cancerous and non-cancerous cells, various deep learning networks were trained, and their performances were evaluated. Among these, the ViT-based model demonstrated superior performance in both patch-level and WSI-level classifications. Furthermore, we proposed an ensemble model as our second-best performing model. Although CNNs are predominantly used in image classification, our methodology demonstrates that ensembling two CNNs can enhance model performance. A notable challenge encountered was the low number of lobular carcinoma cases in the dataset. This was addressed through the augmentation of patches to artificially increase the data. However, the inclusion of a greater number of lobular carcinoma cases in the dataset could further improve the patch-based performance of the models. Looking ahead, the automatic identification of breast cancer and its subtypes holds the potential to facilitate the application of tailored treatment approaches for each specific subtype. This advancement would enable more personalized and effective treatment strategies, improving patient outcomes and advancing the field of breast cancer diagnosis and treatment.

Author contributions Ishrat Jahan contributed to Conceptualization, Methodology, Software, Writing—original draft, and Writing—review and editing. Muhammad E.H. Chowdhury contributed to Conceptualization, Methodology, Supervision, Writing—original draft, and Writing—review and editing. Semir Vranic contributed to Validation, Methodology, Software, and Writing—review and editing. Rafif Mahmood Al Saady contributed to Validation, Methodology, Software, and Writing—review and editing. Saidul Kabir contributed to Conceptualization, Methodology, Software, and Validation. Zahid Hasan Pranto contributed to Conceptualization, Writing—original draft, and Writing—review and editing. Sabiha Jahan Mim contributed to Writing—original draft and Writing—review and editing. Sadia Farhana Nobi contributed to Conceptualization, Methodology, Software, and Validation.

Funding Open Access funding provided by the Qatar National Library. No funding was available for this study. Acknowledgement: The open access publication cost is supported by Qatar National Library (QNL).

Data availability The processed dataset used in this study can be made available upon a reasonable request to the corresponding author.

Declarations

Conflict of interest Authors have no conflict of interest to declare.

Informed consent Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

1. Barbosa AM, Martel F (2020) Targeting glucose transporters for breast cancer therapy: the effect of natural and synthetic compounds. *Cancers* 12(1):154
2. Bardou D, Zhang K, Ahmad SM (2018) Classification of breast cancer based on histology images using convolutional neural networks. *Ieee Access* 6:24680–24693
3. Breast Tumours: WHO Classification of Tumours (2019) (W. C. o. T. E. Board Ed. 5th ed. Vol. 2): World Health Organization
4. Chen H, Wang Y, Guo T, Xu C, Deng Y, Liu Z, Gao W (2021) Pre-trained image processing transformer. Paper presented at the Proceedings of the IEEE/CVF conference on computer vision and pattern recognition
5. Chowdhury MEH, Rahman T, Khandakar A, Mazhar R, Kadir MA, Mahbub ZB, Islam KR, Khan MS, Iqbal A, Emadi NA, Reaz MBI, Islam MT (2020) Can AI help in screening viral and COVID-19 pneumonia? *IEEE Access* 8:132665–132676. <https://doi.org/10.1109/ACCESS.2020.3010287>

6. Chowdhury MH, Chowdhury ME, Khan MS, Ullah MA, Mahmud S, Khandakar A, Hasan A (2023) Self-attention MHDNet: A novel deep learning model for the detection of R-peaks in the electrocardiogram signals corrupted with Magnetohydrodynamic effect. *Bioengineering* 10(5):542
7. Conceicao R (2010) The development of ultra wideband scanning techniques for detection and classification of breast cancer
8. Cutuli B, De Lafontan B, Kirova Y, Auvray H, Tallet A, Avigdor S, Delva C (2015) Lobular carcinoma in situ (LCIS) of the breast: is long-term outcome similar to ductal carcinoma in situ (DCIS)? Analysis of 200 cases. *Radiation Oncol* 10:1–7
9. Dai Y, Gao Y, Liu F (2021) Transmed: transformers advance multi-modal medical image classification. *Diagnostics* 11(8):1384
10. Das K, Conjeti S, Chatterjee J, Sheet D (2020) Detection of breast cancer from whole slide histopathological images using deep multiple instance CNN. *IEEE Access* 8:213502–213511
11. DePolo J (2024) Invasive lobular carcinoma (ILC). Retrieved from <https://www.breastcancer.org/types/invasive-lobular-carcinoma>
12. DeWit SC, Stromberg H, Dallred C (2016) Medical-surgical nursing: concepts & practice: Elsevier Health Sciences
13. Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Gelly S (2020) An image is worth 16x16 words: transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*
14. Faisal MAA, Chowdhury ME, Mahbub ZB, Pedersen S, Ahmed MU, Khandakar A, Bhuiyan EH (2023) NDDNet: A deep learning model for predicting neurodegenerative diseases from gait pattern. *Appl Intell* 53(17):20034–20046
15. Gurcan MN, Boucheron LE, Can A, Madabhushi A, Rajpoot NM, Yener B (2009) Histopathological image analysis: A review. *IEEE Rev Biomed Eng* 2:147–171
16. Hajiabadi H, Babaiyan V, Zabihzadeh D, Hajiabadi M (2020) Combination of loss functions for robust breast cancer prediction. *Comput Electr Eng* 84:106624
17. Hardin PJ, Shumway JM (1997) Statistical significance and normalized confusion matrices. *Photogramm Eng Remote Sens* 63(6):735–739
18. Harmer V (2008) Breast cancer. Part 1: awareness and common benign diseases. *Br J Nursing* 17(15):950–955
19. Hilary Parker AAS (2024). Invasive ductal carcinoma (IDC) and ductal carcinoma in situ (DCIS). Retrieved from <https://www.webmd.com/breast-cancer/ductal-carcinoma-invasive-in-situ#1-2>
20. Hirra I, Ahmad M, Hussain A, Ashraf MU, Saeed IA, Qadri SF, Alfakeeh AS (2021) Breast cancer classification from histopathological images using patch-based deep learning modeling. *IEEE Access* 9:24273–24287
21. Howard AG, Zhu M, Chen B, Kalenichenko D, Wang W, Weyand T, Adam H (2017) Mobilenets: efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*
22. Huang G, Liu Z, Van Der Maaten L, Weinberger KQ (2017) Densely connected convolutional networks. Paper presented at the proceedings of the IEEE conference on computer vision and pattern recognition
23. Janowczyk A, Madabhushi A (2016) Deep learning for digital pathology image analysis: a comprehensive tutorial with selected use cases. *J Pathol Inf* 7(1):29
24. Juppert Q, De Martino F, Marcandalli E, Weigert M, Burri O, Unser M, Sage D (2021) Deep learning enables individual xenograft cell classification in histological images by analysis of contextual features. *J Mamm Gland Biol Neoplasia* 26(2):101–112
25. Kabir S, Vranic S, Al Saady RM, Khan MS, Sarmun R, Alqahtani A, Chowdhury ME (2024) The utility of a deep learning-based approach in Her-2/neu assessment in breast cancer. *Expert Syst Appl* 238:122051
26. Lashof JC, Henderson IC, Nass SJ (2001) Mammography and beyond: developing technologies for the early detection of breast cancer
27. Li C, Lin X, Mao Y, Lin W, Qi Q, Ding X, Yu Y (2022) Domain generalization on medical imaging classification using episodic training with task augmentation. *Comput Biol Med* 141:105144
28. Li C, Liu X, Li W, Wang C, Liu H, Yuan Y (2024) U-KAN Makes Strong Backbone for Medical Image Segmentation and Generation. *arXiv preprint arXiv:2406.02918*
29. Li Y, Wu J, Wu Q (2019) Classification of breast cancer histology images using multi-size and discriminative patches based on deep learning. *IEEE Access* 7:21400–21408
30. Maurício J, Domingues I, Bernardino J (2023) Comparing vision transformers and convolutional neural networks for image classification: a literature review. *Appl Sci* 13(9):5521
31. Mi W, Li J, Guo Y, Ren X, Liang Z, Zhang T, Zou H (2021) Deep learning-based multi-class classification of breast digital pathology images. *Cancer Manag Res* 13:4605–4617. <https://doi.org/10.2147/CMAR.S312608>
32. Mohammed MA, Al-Khateeb B, Rashid AN, Ibrahim DA, Abd Ghani MK, Mostafa SA (2018) Neural network and multi-fractal dimension features for breast cancer classification from ultrasound images. *Comput Electr Eng* 70:871–882
33. Nazeri K, Aminpour A, Ebrahimi M (2018) Two-stage convolutional neural network for breast cancer histology image classification. Paper presented at the Image Analysis and Recognition: 15th International Conference, ICIAR 2018, Póvoa de Varzim, Portugal, June 27–29, 2018, Proceedings 15
34. Obayya M, Maashi MS, Nemri N, Mohsen H, Motwakel A, Osman AE, Alsaid MI (2023) Hyperparameter optimizer with deep learning-based decision-support systems for histopathological breast cancer diagnosis. *Cancers* 15(3):885
35. Pöllänen I, Braithwaite B, Ikonen T, Niska H, Haataja K, Toivanen P, Tolonen T (2014) Computer-aided breast cancer histopathological diagnosis: Comparative analysis of three DTOCS-based features: SW-DTOCS, SW-WDTOCS and SW-3–4-DTOCS. Paper presented at the 2014 4th International Conference on Image Processing Theory, Tools and Applications (IPTA)
36. Raghu M, Unterthiner T, Kornblith S, Zhang C, Dosovitskiy A (2021) Do vision transformers see like convolutional neural networks? *Adv Neural Inf Process Syst* 34:12116–12128
37. Rajpoot N (2019) Her2 Scoring Contest @ Nottingham Pathology 2016. Retrieved from https://warwick.ac.uk/fac/cross_fac/tia/data/her2contest/
38. Reed AEM, Kutasovic JR, Lakhani SR, Simpson PT (2015) Invasive lobular carcinoma of the breast: morphology, biomarkers and omics. *Breast Cancer Res* 17:1–11
39. Roy K, Banik D, Bhattacharjee D, Nasipuri M (2019) Patch-based system for classification of breast histology images using deep learning. *Comput Med Imaging Graph* 71:90–103
40. Salvatorelli L, Puzzo L, Vecchio GM, Caltabiano R, Virzì V, Magro G (2020) Ductal carcinoma in situ of the breast: an update with emphasis on radiological and morphological features as predictive prognostic factors. *Cancers* 12(3):609
41. Sandler M, Howard A, Zhu M, Zhmoginov A, Chen L-C (2018) *Mobilenetv2: Inverted residuals and linear bottlenecks*. Paper presented at the Proceedings of the IEEE conference on computer vision and pattern recognition
42. Sarvamangala D, Kulkarni RV (2022) Convolutional neural networks in medical image understanding: a survey. *Evol Intel* 15(1):1–22
43. Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D (2017) Grad-cam: Visual explanations from deep networks via

- gradient-based localization. Paper presented at the Proceedings of the IEEE international conference on computer vision
44. Shahidi F, Daud SM, Abas H, Ahmad NA, Maarop N (2020) Breast cancer classification using deep learning approaches and histopathology image: a comparison study. *IEEE Access* 8:187531–187552
 45. Siegel RL, Miller KD, Wagle NS, Jemal A (2023) Cancer statistics, 2023. *Ca Cancer J Clin* 73(1):17–48
 46. Smolarz B, Nowak AZ, Romanowicz H (2022) Breast cancer—epidemiology, classification, pathogenesis and treatment (review of literature). *Cancers* 14(10):2569
 47. Sohail A, Khan A, Wahab N, Zameer A, Khan S (2021) A multi-phase deep CNN based mitosis detection framework for breast cancer histopathological images. *Sci Rep* 11(1):6215
 48. Sokolova M, Lapalme G (2009) A systematic analysis of performance measures for classification tasks. *Inf Process Manage* 45(4):427–437
 49. Srikanthamurthy MM, Rallabandi VS, Dudekula DB, Natarajan S, Park J (2023) Classification of benign and malignant subtypes of breast cancer histopathology imaging using hybrid CNN-LSTM based transfer learning. *BMC Med Imaging* 23(1):19
 50. Sun L, Li C, Ding X, Huang Y, Chen Z, Wang G, Paisley J (2022) Few-shot medical image segmentation using a global correlation network with discriminative embedding. *Comput Biol Med* 140:105067
 51. Teng Q, Liu Z, Song Y, Han K, Lu Y (2022) A survey on the interpretability of deep learning in medical diagnosis. *Multimedia Syst* 28(6):2335–2355
 52. Touvron H, Cord M, Douze M, Massa F, Sablayrolles A, Jégou H (2021) Training data-efficient image transformers & distillation through attention. Paper presented at the International conference on machine learning
 53. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Polosukhin I (2017) Attention is all you need. *Adv Neural Inf Process Syst*, 30
 54. Veta M, Pluim JP, Van Diest PJ, Viergever MA (2014) Breast cancer histopathology image analysis: a review. *IEEE Trans Biomed Eng* 61(5):1400–1411
 55. Wang H, Wang Z, Du M, Yang F, Zhang Z, Ding S, Hu X (2020) Score-CAM: Score-weighted visual explanations for convolutional neural networks. Paper presented at the Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops
 56. Weinberg RA (1996) How cancer arises. *Sci Am* 275(3):62–70
 57. Won JR, Gao D, Chow C, Cheng J, Lau SY, Ellis MJ, Nielsen TO (2013) A survey of immunohistochemical biomarkers for basal-like breast cancer against a gene expression profile gold standard. *Modern Pathol* 26(11):1438–1450
 58. Yang Q, Li W, Li B, Yuan Y (2023) Mrm: Masked relation modeling for medical image pre-training with genetics. Paper presented at the Proceedings of the IEEE/CVF International Conference on Computer Vision
 59. Zhao H (2021) The prognosis of invasive ductal carcinoma, lobular carcinoma and mixed ductal and lobular carcinoma according to molecular subtypes of the breast. *Breast Cancer* 28:187–195
 60. Zhou Y, Zhang C, Gao S (2022) Breast cancer classification from histopathological images using resolution adaptive network. *IEEE Access* 10:35977–35991

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Ishrat Jahan¹ · Muhammad E. H. Chowdhury²  · Semir Vranic³ · Rafif Mahmood Al Saady³ · Saidul Kabir⁴ · Zahid Hasan Pranto¹ · Sabiha Jahan Mim¹ · Sadia Farhana Nobi²

✉ Muhammad E. H. Chowdhury
mchowdhury@qu.edu.qa

✉ Rafif Mahmood Al Saady
rafif@qu.edu.qa

Ishrat Jahan
i.jahan1533@gmail.com

Semir Vranic
svranic@qu.edu.qa

Saidul Kabir
saidul-2016614965@eee.du.ac.bd

Zahid Hasan Pranto
zpranto96@gmail.com

Sabiha Jahan Mim
mimsabiha59@gmail.com

Sadia Farhana Nobi
dr.sadianobi@gmail.com

¹ Department of Computer Science and Engineering, Shahjalal University of Science and Technology, Sylhet, Bangladesh

² Department of Electrical Engineering, Qatar University, 2713, Doha, Qatar

³ College of Medicine, QU Health, Qatar University, Doha, Qatar

⁴ Department of Electrical and Electronic Engineering, University of Dhaka, Dhaka 1000, Bangladesh